Shivam

Academic CV

Junior Research Fellow (JRF)
Theoretical Condensed Matter Physics
Physical Research Laboratory (PRL), Ahmedabad, India

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Research Profile

Researcher in theoretical condensed matter physics interested in unconventional magnetism, altermagnetism, momentum-dependent spin splitting, and symmetry-driven phenomena in quantum materials. My academic interests include understanding the relationship between crystal symmetries, magnetic order, electronic band structures, and topological phases of matter.

Research Interests

- *Unconventional magnetism and altermagnetism*
- *Momentum-dependent spin splitting and spin textures*
- *Symmetry analysis of magnetic and electronic structures*
- *Topological phases of matter and topological insulators*
- *Wannier centres and topological charge pumping*
- *Magnetism, spintronics, and superconductivity*

Education

Ph.D. in Physics [2025–Present]
Physical Research Laboratory (PRL), Ahmedabad, India
Research area: Theoretical Condensed Matter Physics
Supervisor: Dr. Paramita Dutta

M.Sc. in Physics [2022–2024]
University of Allahabad
Dissertation/Project: Chiral Symmetry Breaking in Quark–Hadron Phase Transition

B.Sc. in Physics [2019–2022]
University of Allahabad

Research Experience

Junior Research Fellow (JRF) [2025–Present]
Physical Research Laboratory, Ahmedabad, India

- *Study unconventional magnetic orders, with particular emphasis on altermagnetism and momentum-dependent spin splitting.*
- *Investigate the role of crystal and magnetic symmetries in determining spin-resolved electronic band structures.*
- *Explore minimal theoretical models for different momentum- dependent magnetic orders, including p-wave, d-wave, and f-wave forms.*
- *Study topological phases, including the BHZ and Kane–Mele models, and their associated symmetry properties.*

Academic and Research Projects

1. Unconventional Magnetism and Altermagnetism

- *Study symmetry-based descriptions of unconventional magnetic orders and their momentum-dependent spin splitting.*
- *Explore minimal model Hamiltonians and spin-resolved electronic structures for different magnetic symmetries.*

2. Introduction to Topological Phases in Condensed Matter

- *Study topological insulators, Chern insulators, and quantum spin Hall systems.*
- *Investigate the BHZ model, Kane–Mele model, chiral symmetry, and topological invariants.*

3. Topological Phases in One-Dimensional Systems

- *Study winding numbers, Wannier centres, and topological charge pumping in one-dimensional systems.*

Seminars and Presentations

- **Doctoral Student Committee (DSC) Seminar:** *"Unconventional Magnetism: From Symmetry to Momentum-Dependent Spin Splitting." Physical Research Laboratory, Ahmedabad.* [2026]
- **Academic Project Presentation:** *"Introduction to Topological Phases in Condensed Matter."* [2026]

Technical Skills

Programming: *Python, NumPy, Matplotlib*
Scientific Computing: *Numerical calculations and visualization of electronic dispersions and Fermi surfaces*
Typesetting: *LaTeX, Beamer, LibreOffice*
Operating Systems: *Windows, Linux, Ubuntu*
Other Tools: *VS Code, Jupyter Notebook, Git/GitHub*

Academic Service and Activities

- *Member, PRL Student's Committee, Physical Research Laboratory, Ahmedabad.* [2026–Present]
- *Volunteer, Hostel Cultural and Maintenance Committee, Physical Research Laboratory.* [2025–2026]

Publications

- to be updated

Awards and Qualifications

- *GATE–2025/26 and JEST–2025 with AIR–107*
- *Inspire Fellowship Awardee, PRL–JRF Fellowship*

References

To be Updated